



Daksh Agarwal

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B.Tech - Computer Science Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	9.63 (Current)	2023-Present
Senior Secondary	CBSE Board	95.8%	2023
Secondary	CBSE Board	97.8%	2021

EXPERIENCE

- Research Consultant** May 2025 - Present
WorldQuant LLC
 - Developed systematic **trading algorithms** (alphas) using quantitative research to identify and exploit market inefficiencies.
 - Submitted alphas were backtested on **10 years of US equity data**, evaluated by **Sharpe Ratio**, **Turnover** and **Returns**.
 - Presently developed **100+ alpha strategies**, some having sharpe ratio exceeding 1.5 and pairwise correlation less than 0.7.

PROJECTS

- Curating Alphas on BTC/USDT and ETH/USDT Crypto Markets** December 2024
Inter IIT Tech Meet 13.0, IIT Bombay - Gold bit.ly/untrade-strategy
 - Developed a **Q-learning** based cryptocurrency trading strategy, optimizing BTC trades in highly volatile markets.
 - Enhanced signal clarity using **wavelet transforms**, **Kalman filters**, and **Heikin Ashi candles** for trend identification.
 - Achieved a profit margin of **448.06%** with **157 trades**, a Sharpe Ratio of **7.80**, and a maximum drawdown of **22.34%**
 - Designed a **correlation** based ETH trading framework leveraging BTC-ETH divergence with multi-confirmation.
- Optimization for Containment Field Construction** January 2025
Inter Hostel Technical Competition, Kriti 2025 - Bronze bit.ly/kriti-opti
 - Developed and implemented a **greedy algorithm** to solve a strictly time-constrained computational geometry challenge.
 - Applied **clustering heuristics** to select the top 150 clusters and eliminate the lowest-performing ones for optimal results.
 - Created an **ensemble of four competing heuristics**, selecting the best-performing result for each test case.
 - Optimized **data processing** to handle 10,000-point datasets efficiently, meeting a **10-minute time limit** for 20 test cases.
- Deep Learning-Based Portfolio Optimization** February 2025
Finance and Economics Club, IIT Guwahati bit.ly/portfolio-DL
 - Built an end-to-end **LSTM network** to directly maximize a portfolio's Sharpe ratio, bypassing traditional forecasting steps.
 - Validated the strategy's robustness over a period of 10 years, including the **2020 COVID-19 market crash**.
 - Delivered superior **risk-adjusted returns**, attaining a **Sharpe Ratio** of **1.96** vs baseline's **1.58** under identical scenarios.

ACHIEVEMENTS

- Inter IIT Tech Meet 13.0**: Secured the **Gold Medal** in the Zelta-Automations Quant Strategy Challenge. 2024
- Joint Entrance Examination Mains**: Secured **All India Rank 623** among 14,00,000+ students. 2023
- Joint Entrance Examination Advanced**: Secured **All India Rank 1698** among 1,50,000+ students. 2023
- IICPC QuantFest 2025**: Secured **All India Rank 429** in Prelims among 10,000+ registrants. 2025
- Codeforces**: Maximum rating - **1551** | **Specialist** | Handle - *hskad*
 - Codeforces round 1032 (Division 3): Secured **Global Rank 651** among 37,000+ participants.
- CodeChef**: Maximum rating - **1865** | **4 Stars** | Handle - *alt_hskad*
 - Starters 186 Division 4: Secured a **Global Rank of 98** among 14,000+ participants.
 - Starters 190 Division 2: Secured a **Global Rank of 266** among 12,000+ participants.
- Kriti - Inter Hostel Optimization Challenge**: Secured **Rank 3** in the Geometric Optimization Challenge. 2025
- Secured a branch change**: Achieved a branch change from **ECE to CSE** through exceptional academic performance. 2024
- National Talent Search Examination (NTSE)**: Secured a **state rank of 70** in the first stage. 2021

TECHNICAL SKILLS

- Programming**: C/C++, Python*, JavaScript*
- Libraries**: NumPy, Pandas, Matplotlib, Plotly
- Web Technologies**: HTML, CSS, React.js, Node.js, Express.js*
- Miscellaneous**: Git, MySQL, SystemVerilog*, LaTeX*

KEY COURSES TAKEN

- Computer Science**: Data Structures and Algorithms, Design and Analysis of Algorithms*, Database Management Systems*, Digital Design*, Computer Architecture and Organization, Formal Languages, Automata Theory and Computation*
- Mathematics**: Probability Theory and Random Processes, Number Theory and Abstract Algebra, Discrete Mathematics*
- Finance and Economics**: Introductory Options, Introductory Macroeconomics

*Awarded AA/AS grade (10/10)

POSITIONS OF RESPONSIBILITY

- Quant Head, Finance and Economics Club, IIT Guwahati** April 2025 - Present
 - Designed a comprehensive course on **Futures, Options and other Derivatives**, engaging over 1500+ student participants.
 - Guided a **quant summer-project program**, providing technical mentorship to over 400+ student participants.